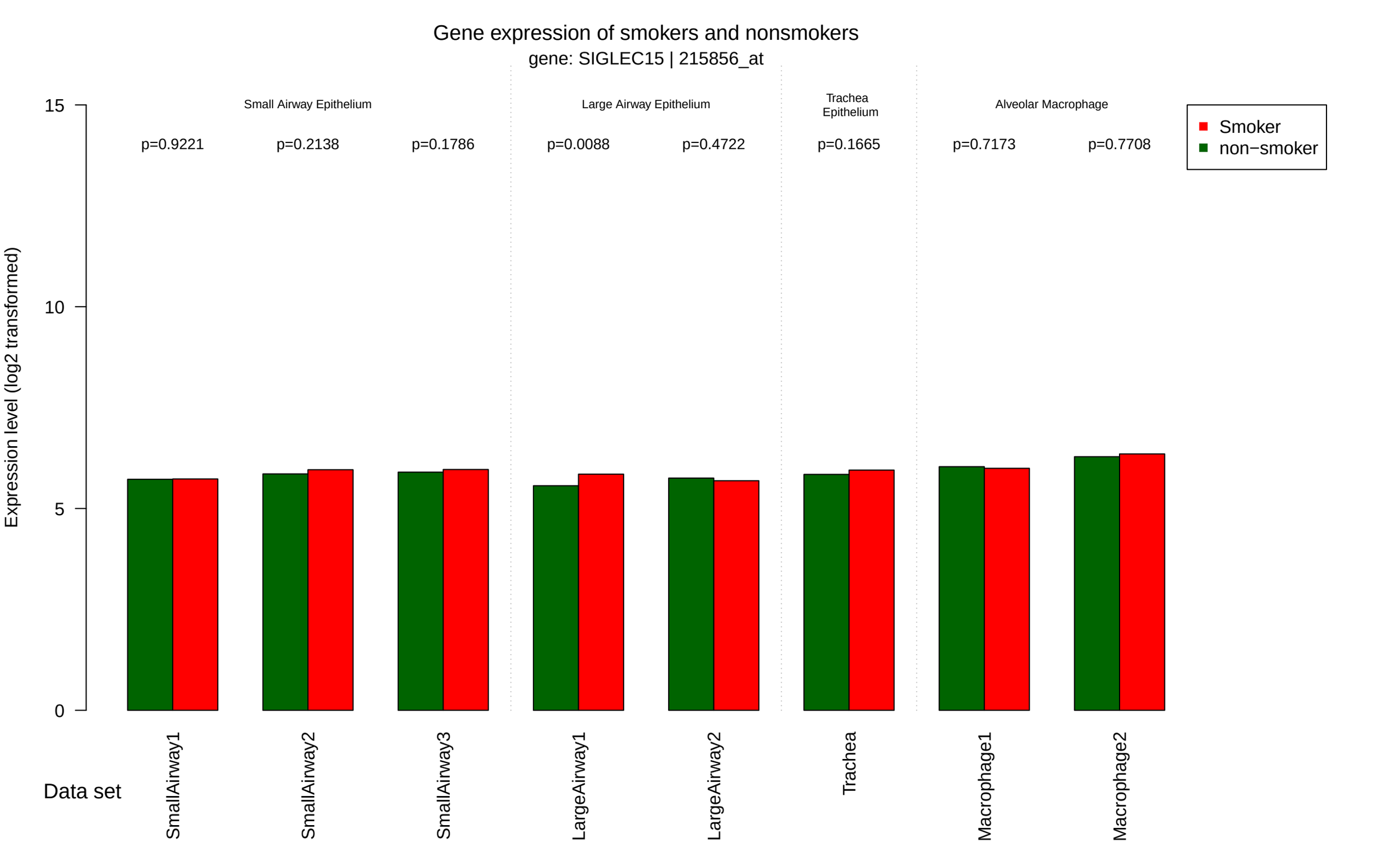
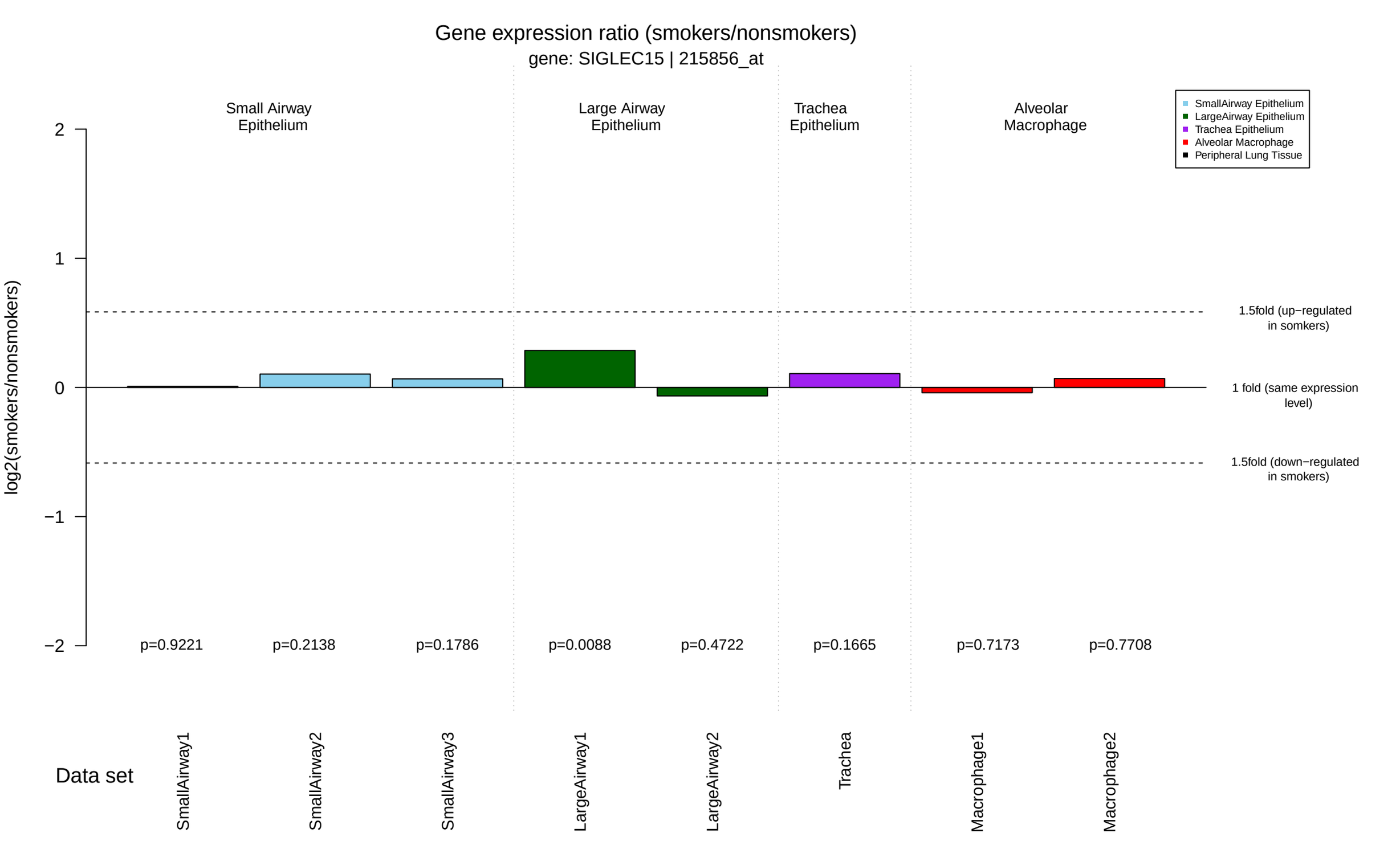
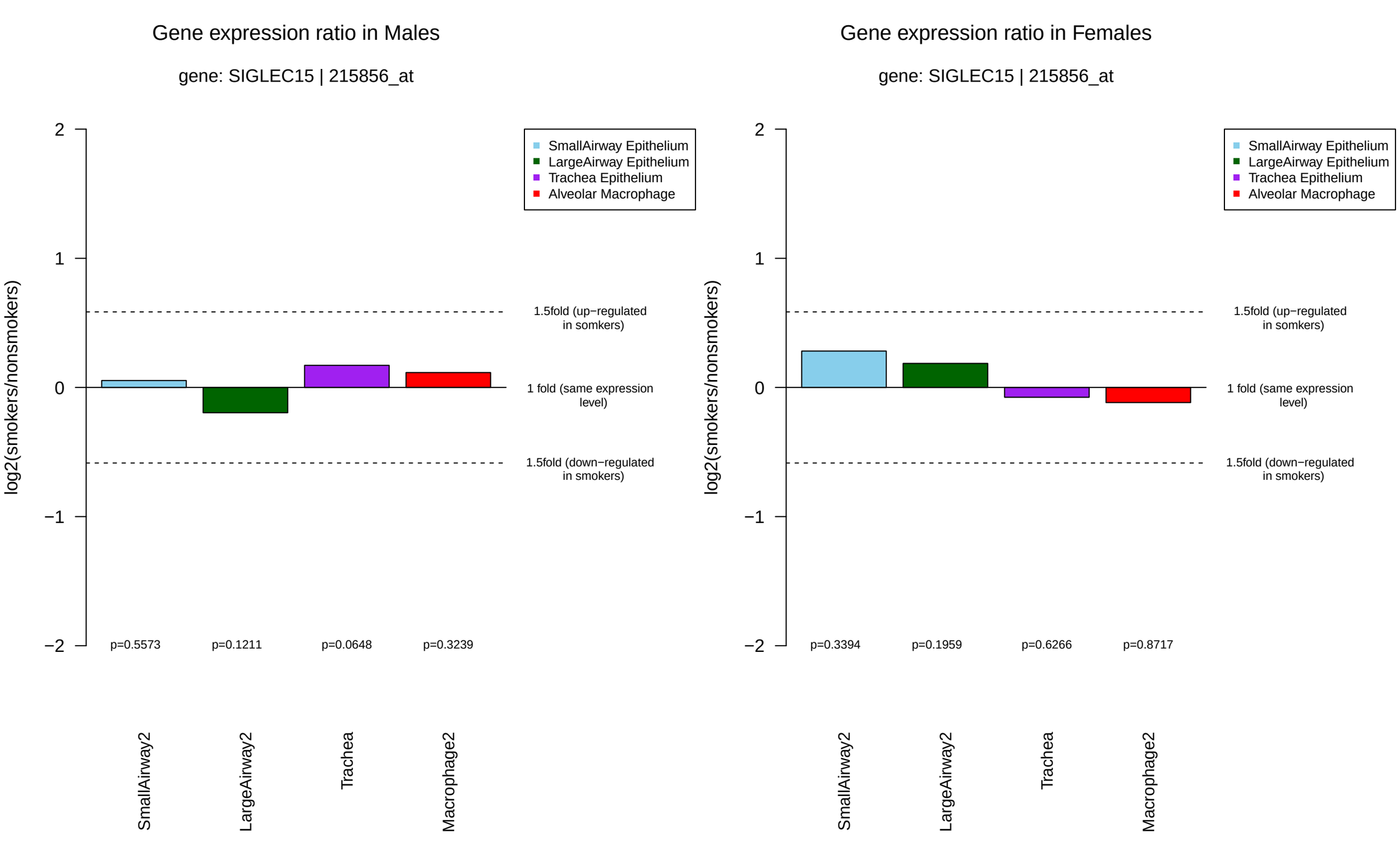
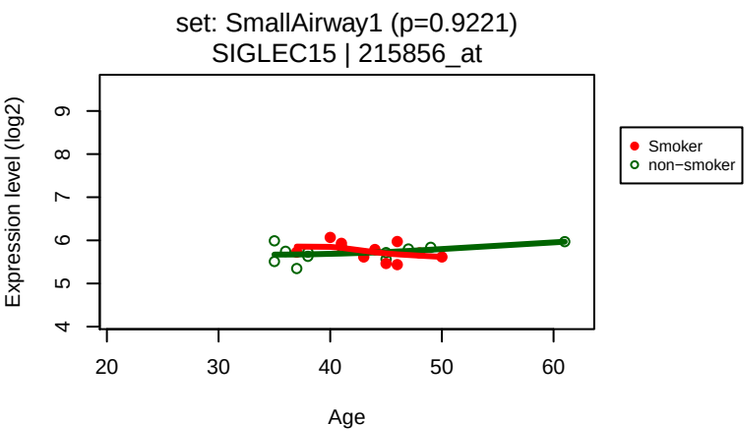
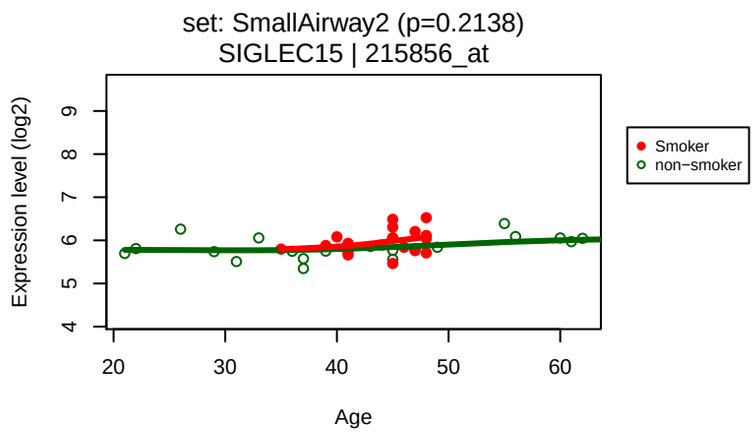
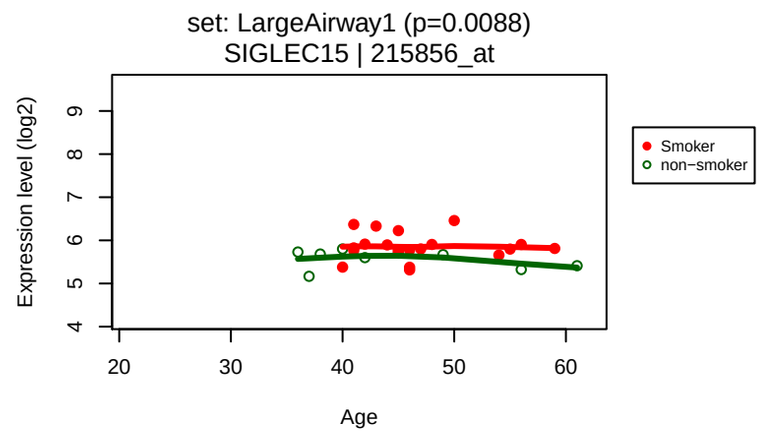

Gene expression of smokers and nonsmokers
gene: SIGLEC15 | 215856_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9221
p=0.2138
p=0.1786
p=0.0088
p=0.4722
p=0.1665
p=0.7173
p=0.7708
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

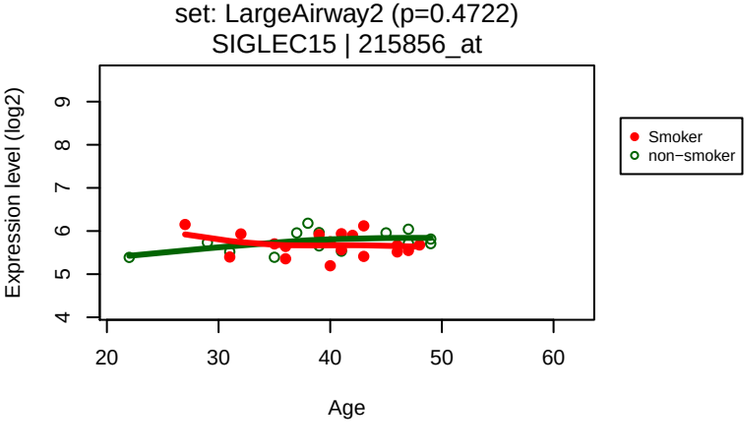

Gene expression ratio (smokers/nonsmokers)
gene: SIGLEC15 | 215856_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9221
p=0.2138
p=0.1786
p=0.0088
p=0.4722
p=0.1665
p=0.7173
p=0.7708
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

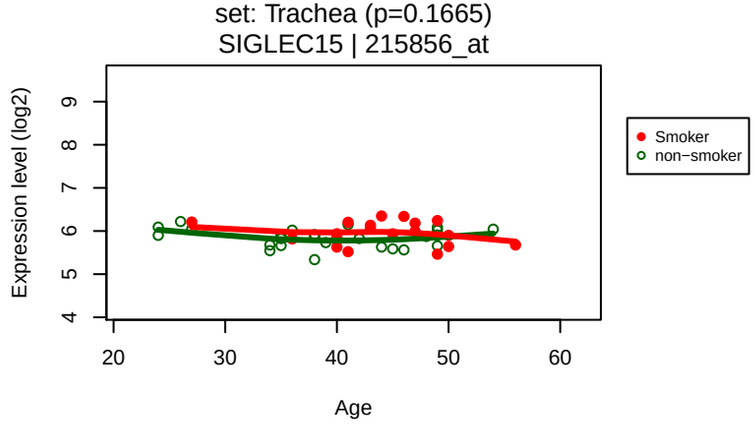

Gene expression ratio in Males
gene: SIGLEC15 | 215856_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5573
p=0.1211
p=0.0648
p=0.3239
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIGLEC15 | 215856_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3394
p=0.1959
p=0.6266
p=0.8717
SmallAirway2
LargeAirway2
Trachea
Macrophage2

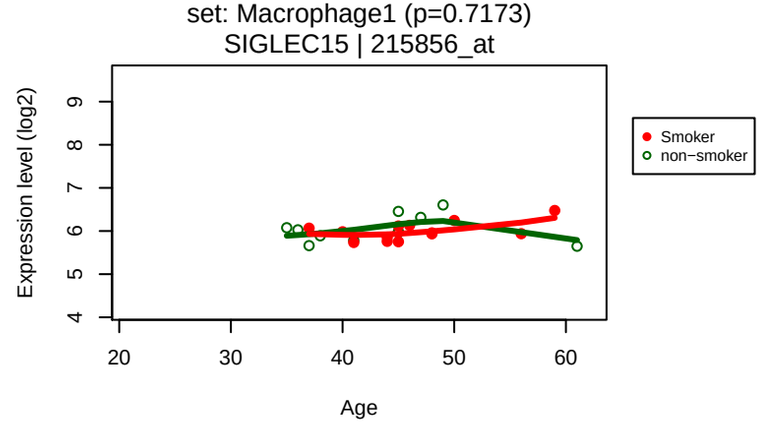

set: SmallAirway1 (p=0.9221)
SIGLEC15 | 215856_at
Expression level (log2)
Age
Smoker
non−smoker

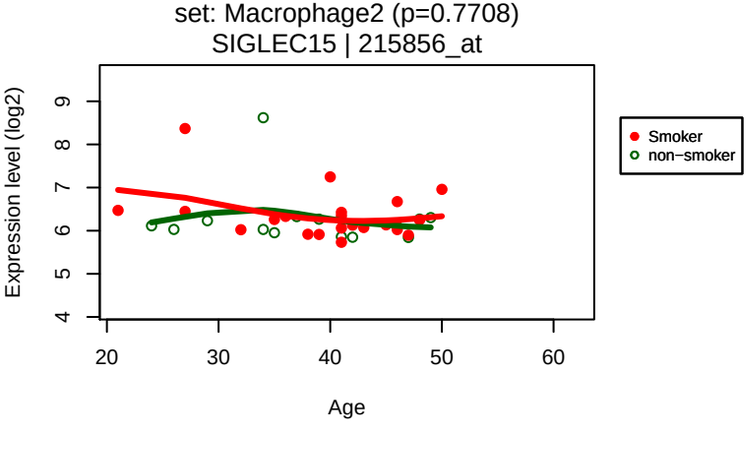

set: SmallAirway2 (p=0.2138)
SIGLEC15 | 215856_at
Expression level (log2)
Age
Smoker
non−smoker

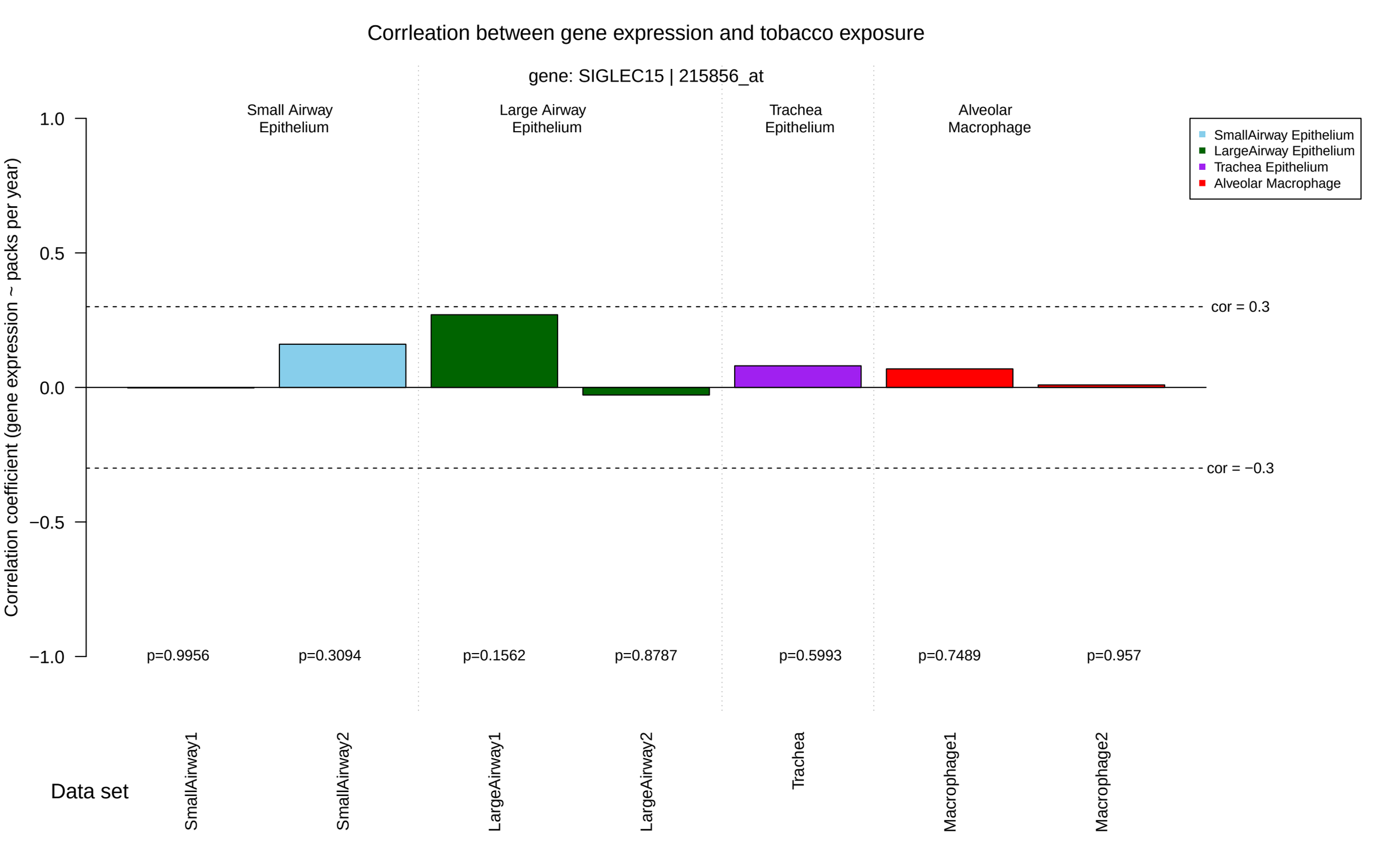

set: LargeAirway1 (p=0.0088)
SIGLEC15 | 215856_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4722)
SIGLEC15 | 215856_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1665)
SIGLEC15 | 215856_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7173)
SIGLEC15 | 215856_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7708)
SIGLEC15 | 215856_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIGLEC15 | 215856_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9956
p=0.3094
p=0.1562
p=0.8787
p=0.5993
p=0.7489
p=0.957
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2